SUDARSHAN V

BE ECE AU-CEG

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ABSTRACT

Time Division Multiple Access (TDMA) is a channel access technique in which a shared wireless medium is divided into recurring time slots. In a multi-hop wireless network, the topology determines how nodes connect to a gateway, how many relays are required, how far the network can extend, and how efficiently transmissions can be scheduled. This project studies a software-based topology optimization workflow that evaluates candidate network structures using frame length, hop count, network reach, covered area and area efficiency.

The supplied experimental archive contains a 19-radio baseline with a 550 m communication range, a 476-radio case with a 422 m range, one additional 50-radio scalability case, and five objective-ranking reruns for the 476-radio case. The 476radio evaluation reports 13 feasible candidate topologies. Multiple spider layouts achieve the theoretical lower bound of 475 slots, while the chain topology provides the greatest reach of 180.405 km at the cost of a 1,422-slot frame. The four-arm spider achieves the highest reported area-per-slot efficiency of 0.3101 km²/slot. For the 19radio case, the three-arm spider achieves 0.5778 km²/slot with an 18-slot frame.

The results show that no single topology is optimal for every performance objective. Topology selection must therefore be based on the intended application: low frame length, low hop count, maximum reach or area efficiency. The report also documents the evidence boundary: the supplied batch includes only one intermediate scalability point and does not contain completed traffic-load, interference, minimum-spacing, link-fraction or multi-seed sweeps. Those analyses are identified as future work rather than reported as completed experiments.

Keywords: TDMA, wireless mesh network, topology optimization, multi-hop routing, frame length, hop count, network coverage, scheduling.

LIST OF ABBREVIATIONS

Abbreviation	Meaning
TDMA	Time Division Multiple Access
MAC	Medium Access Control
GW	Gateway
RF	Radio Frequency
CSV	Comma-Separated Values
JSON	JavaScript Object Notation
WSL2	Windows Subsystem for Linux 2
N	Total number of radios
H	Hop count
S	Number of TDMA slots in a frame
SINR	Signal-to-Interference-plus-Noise Ratio
DSATUR	Degree of Saturation (graph colouring heuristic)

1. INTRODUCTION

1.1 Background

Wireless multi-hop networks are useful when direct communication between every node and a central gateway is not possible or desirable. Nodes outside the gateway's direct radio range forward their data through intermediate nodes. This increases the effective network footprint, but it also introduces routing depth, relay load and scheduling constraints.

In a shared wireless channel, simultaneous transmissions may interfere. TDMA addresses channel contention by assigning transmissions to specific time slots. A well-designed schedule can reduce collisions and make access predictable. However, the number of slots required depends on the topology and the interference relationships among links.

1.2 Motivation

A topology that maximizes geographical reach may create long paths and large frame lengths. A topology that minimizes hop count may use more parallel branches and may not maximize covered area per slot. The project is motivated by the need to compare these trade-offs quantitatively rather than selecting a topology based only on its visual appearance.

1.3 Project Overview

The software workflow accepts network size and radio parameters, constructs candidate topologies, checks feasibility constraints, generates TDMA schedules and calculates performance metrics. It exports structured summaries and plots that allow the candidate layouts to be compared.

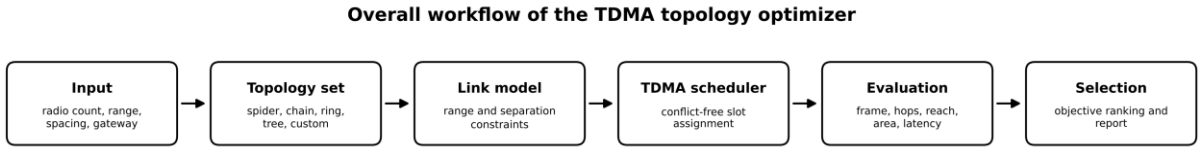


Figure 1.1: Overall workflow of the TDMA topology optimizer.

1.4 Organization of the Report

Chapter 2 introduces the relevant technical concepts. Chapters 3 and 4 define the problem and mathematical model. Chapters 5 and 6 describe the optimization workflow and implementation. Chapters 7 and 8 document the experimental configuration and results. Chapters 9 to 11 discuss the findings, limitations, future work and conclusion.

2. LITERATURE AND TECHNICAL BACKGROUND

2.1 Wireless Multi-hop Networks

A multi-hop network consists of nodes that communicate directly with nearby neighbours. A node that cannot reach the gateway directly forwards its packet through one or more relay nodes. The path length, measured in hops, is an important indicator of routing depth and potential forwarding delay.

2.2 TDMA Medium Access

TDMA divides time into frames, and each frame contains a sequence of slots. A transmission is permitted only in its assigned slot. The schedule must respect the model's conflict constraints so that incompatible transmissions are not assigned to the same slot. The exact conflict definition depends on the radio and interference model used by the implementation.

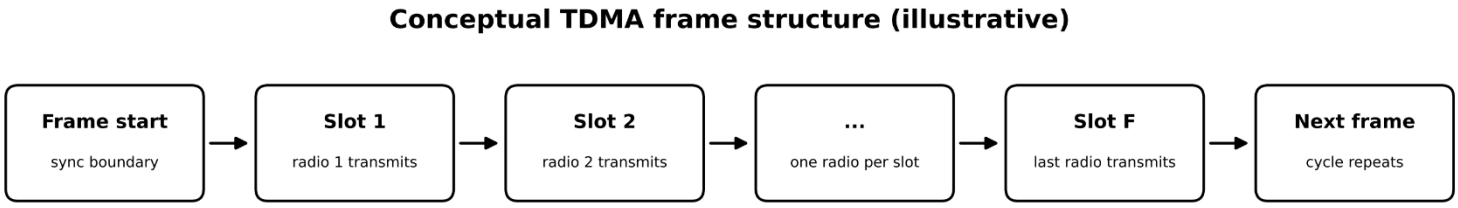
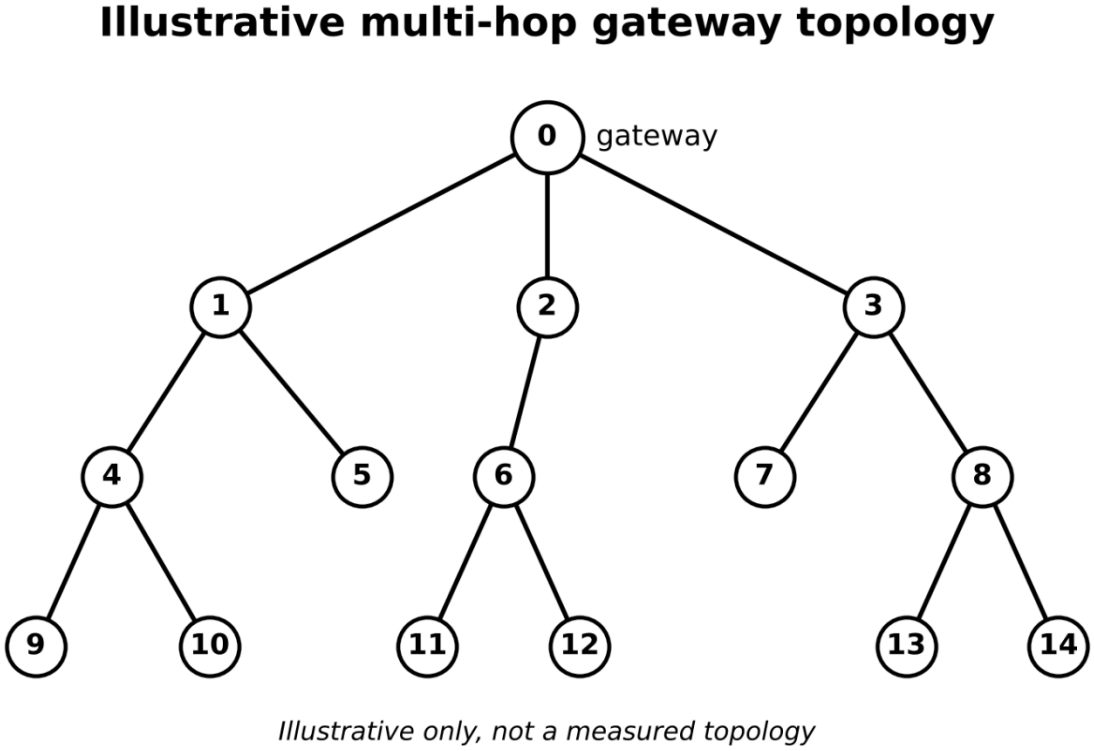


Figure 2.1: Conceptual recurring TDMA frame. This is an explanatory illustration, not an exported schedule.

2.3 Network Topology

A topology defines the connectivity structure between the gateway and the radios. Common structures include a chain, ring, tree and radial or spider-like arrangements. A chain can extend over a long distance but has high hop depth. A radial structure can provide parallel paths from the gateway and reduce path depth, depending on the



number and length of its arms.

Figure 2.2: Illustrative multi-hop gateway topology; not an experimental layout.

2.4 Performance Metrics

Metric	Interpretation	Why it matters
Frame length (slots)	Number of time slots in one repeating frame	Lower values improve schedule compactness
Maximum hops	Largest gateway-to-node path length	Indicates worst-case routing depth
Total hops	Sum of path lengths over all nongateway radios	Represents aggregate forwarding burden
Reach (m)	Farthest reported network extent	Indicates geographical extension
Covered area (km²)	Reported area covered by a candidate layout	Represents spatial footprint

Area per slot (km²/slot)	Covered area divided by frame slots	Combines coverage and schedule cost
Latency bound (slots)	Reported slot-based latency estimate	Indicates delay under stated assumptions

3. PROBLEM DEFINITION AND OBJECTIVES

3.1 Problem Statement

Given a gateway, a set of wireless radios and a set of communication constraints, determine a feasible network topology and TDMA schedule that provide a favourable trade-off among frame length, hop count, reach and covered-area efficiency.

3.2 Objectives

- Generate and evaluate multiple candidate topologies for a given radio count and communication range.
- Check candidate feasibility against link and minimum-separation constraints.
- Construct a TDMA schedule and measure the required frame length.
- Compare candidates using hop count, reach, covered area and area-per-slot efficiency.
- Evaluate different objective rankings to understand how the preferred topology changes.
- Document experimental results and identify the limits of the available test evidence.

3.3 Scope

This report evaluates the software-generated results supplied in the experimental ZIP. It does not claim physical radio measurements, over-the-air validation, energy measurements or a deployed wireless network. The results are simulation/software outputs and should be interpreted within the assumptions of the optimizer.

4. SYSTEM MODEL AND MATHEMATICAL FORMULATION

4.1 Network Representation

Represent the network as a graph G = (V,E), where V is the set of radios including the gateway and E is the set of feasible communication links. Let the gateway be g. Each non-gateway node must have a path to g. A candidate topology is feasible only when its links satisfy the implemented communication and geometric constraints.

4.2 Communication Range

Let R denote the effective communication range. A candidate link between nodes i and j is considered only when its distance is compatible with the configured link model. The supplied 476-radio case uses R = 422 m and a link-distance factor of 0.9. The 19-radio baseline uses R = 550 m. The exact radio propagation environment is not represented by these geometric settings alone.

4.3 Minimum Separation

The minimum-separation setting is a geometric constraint intended to prevent radios from being placed too close to one another. In the 476-radio case, the configured minimum separation is 211 m, equal to one-half of the 422 m range. In the 19-radio baseline, the minimum separation is 275 m, equal to one-half of the 550 m range.

4.4 Frame-Length Lower Bound

For N radios including one gateway, there are N − 1 non-gateway sources. Under the experiment's one-packet-per-source-per-frame demand and single-gateway reception assumption, at least one scheduled source transmission is required for each nongateway radio. Therefore:

$S_{min} \geq N - 1$

For N = 476, $S_{min} \geq 475$ slots. For N = 50, $S_{min} \geq 49$ slots. For N = 19, $S_{min} \geq 18$ slots. This is a lower bound under the stated demand model; it is not a universal bound for every possible traffic pattern or spatial-reuse schedule.

4.5 Hop Count

Let h_i be the number of links on the selected path from node i to the gateway. The maximum hop count is $H_{max} = \max(h_i)$, and total hops are $H_{total} = \Sigma h_i$ over all non-gateway nodes. A lower H_{max} generally reduces routing depth, while H_{total} reflects aggregate path length.

4.6 Coverage Efficiency

The reported area-per-slot metric is defined as:

$\eta_A = A_{covered} / S_{frame}$ where $A_{covered}$ is the reported covered area in km^2 and S_{frame} is the number of slots per frame. A larger value indicates more reported coverage per scheduled slot. This metric should be interpreted together with reach and hop count; it does not independently describe throughput, energy or real-world coverage probability.

4.7 Latency Interpretation

The supplied summaries contain slot-based latency estimates. These values are not seconds because no physical slot duration is specified in the evidence used here. The 476-radio summary also reports a schedule-aware worst-case value of 28,498 slots for the selected four-arm spider. This is distinct from the topology-level hop-based latency bound reported in the candidate summary. They are kept separate in this report to avoid treating different calculations as interchangeable.

Symbol	Description	Unit
N	Total number of radios including gateway	count
R	Effective communication range	m
S	TDMA frame length	slots
h_i	Hop count for node i	hops
H_max	Maximum hop count	hops
H_total	Total hop count	hops
A_covered	Reported covered area	km²
η_A	Area per slot	km²/slot

5. PROPOSED METHODOLOGY

The methodology is organized into candidate generation, feasibility checking, scheduling, metric calculation and objective-based ranking.

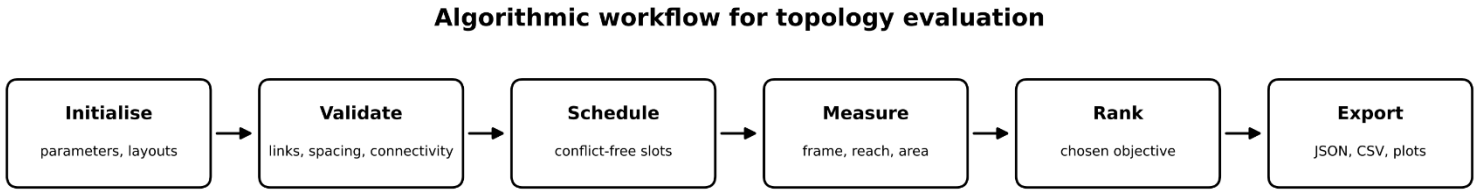


Figure 5.1: Algorithmic workflow used to evaluate candidate topologies.

5.1 Candidate Topology Generation

The result summaries include spider topologies with different arm counts, a chain, a ring, tree candidates and a custom searched layout. The candidate set is evaluated under the same radio-count and radio-parameter configuration for each comparison.

5.2 Feasibility Checking

Candidate layouts are checked for geometric separation and link feasibility. The supplied summaries identify specific rejected candidates, including a 12-arm spider and a star arrangement in the 476-radio case. These failures apply to the generated candidates and configured constraints; they do not prove that all topologies of those broad classes are infeasible.

5.3 TDMA Scheduling

After a candidate is accepted, the scheduler assigns transmission slots subject to the implementation's conflict rules. The resulting frame length is recorded. In the 476radio case, multiple spider candidates attain 475 slots, matching the lower bound derived for the one-packet-per-source model.

5.4 Objective Ranking

The project evaluates different ranking objectives, including frame, latency, reach, area and area-per-slot efficiency. Each objective emphasizes a different network characteristic. The ranking result should therefore be read as the best candidate for that objective, not as a single universally best topology.

5.5 Custom Search

The 476-radio custom search is reported as starting from the four-arm spider and evaluating 2,000 proposals. The final custom candidate has 495 slots and does not outperform the four-arm spider on area-per-slot efficiency. This outcome indicates that the search did not find an improvement in the executed run; it is not a proof of global optimality.

6. SOFTWARE DESIGN AND IMPLEMENTATION:6.1 Execution Environment

Item	Reported environment / project detail
Operating system	Ubuntu 24.04 under WSL2
Python	Python 3.12
Execution	Python virtual environment (.venv)
Main program	tdma_topology_optimizer.py
Result formats	JSON, CSV, Markdown summaries and PNG plots
Project directory	~/tdma_topology_optimizer

6.2 Input Parameters

The command-line interface documented in the project workflow includes controls for ranking objective, traffic load, interference, minimum separation, link-distance factor, random seed, search steps, maximum arms and output directory. The experimental evidence included in the supplied archive does not contain completed sweeps for every parameter.

6.3 Output Files

- Summary JSON: machine-readable experiment configuration and metrics.
- Layout CSV: node coordinates or topology layout data for downstream analysis.
- PNG figures: visual representation of candidate topology and comparisons.
- Markdown summary: compact human-readable result overview.
- Run log: records the command execution and completion status.

6.4 Reproducibility

For repeatability, retain the exact command line, random seed, Python environment, source script version and output directory for each run. The current report is based on the output archive; it does not independently certify the source code's internal implementation beyond the supplied outputs.

7. EXPERIMENTAL SETUP

7.1 Experiment Matrix

Case	Radios (N)	Range	Min. separation	Traffic model	Evidence status
Small baseline	19	550 m	275 m	1 packet per nongateway radio/frame	Completed
Large baseline	476	422 m	211 m	1 packet per nongateway radio/frame	Completed
Intermediate scale	50	422 m	Not independently stated in the summary	As configured in run	Completed
Objective reruns	476	422 m	211 m	Same base case	5 objective rankings completed

7.2 476-Radio Configuration

The large-scale case contains 476 radios including the gateway, resulting in 475 nongateway radios. The configured effective range is 422 m, minimum separation is 211 m, link-distance factor is 0.9 and interference range is 1.5 times the radio range. The gateway packet demand is 475 packets per frame.

7.3 19-Radio Configuration

The small baseline contains 19 radios, including the gateway, with an effective range of 550 m. There are 18 non-gateway radios, so the theoretical frame lower bound under the stated traffic assumption is 18 slots.

7.4 Evidence Inventory

The archive contains baseline outputs for the 19-radio and 476-radio cases, five 476radio objective-ranking reruns, and one 50-radio scalability run. It does not contain completed result folders for the planned traffic-load, interference, minimumseparation, link-fraction and multi-seed sweeps.

8. RESULTS AND ANALYSIS

8.1 Results for the 19-Radio Baseline

Fifteen feasible topologies are reported in the 19-radio summary. Several spider topologies achieve the 18-slot lower bound. The three-arm spider provides the highest reported area-per-slot value, while the chain provides the greatest reach.

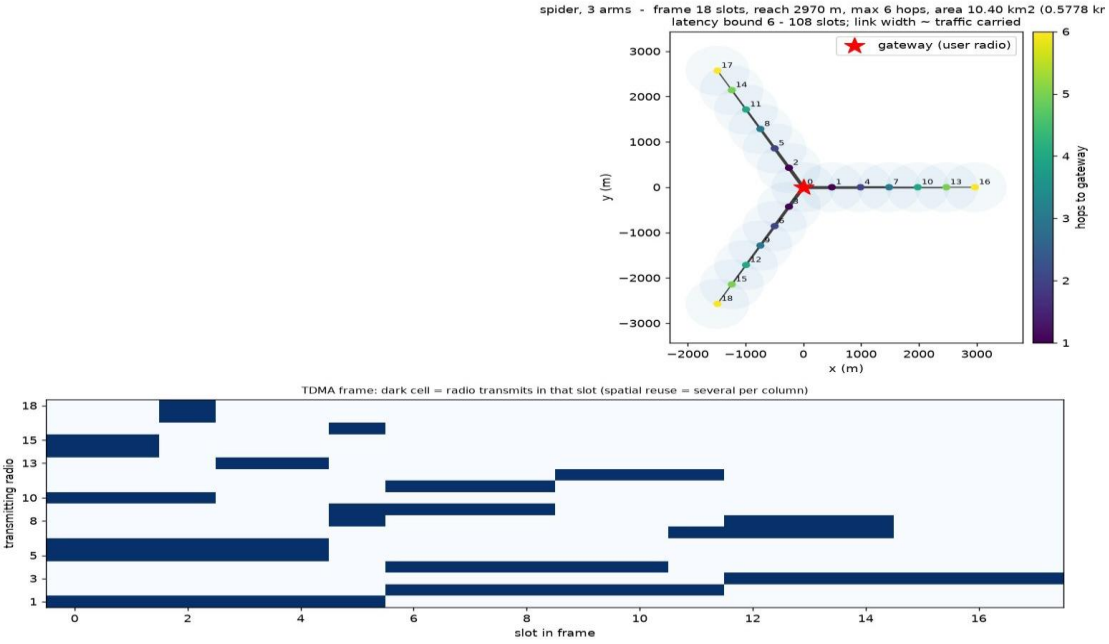


Figure 8.1: Best topology output for the 19-radio baseline (as exported by the optimizer).

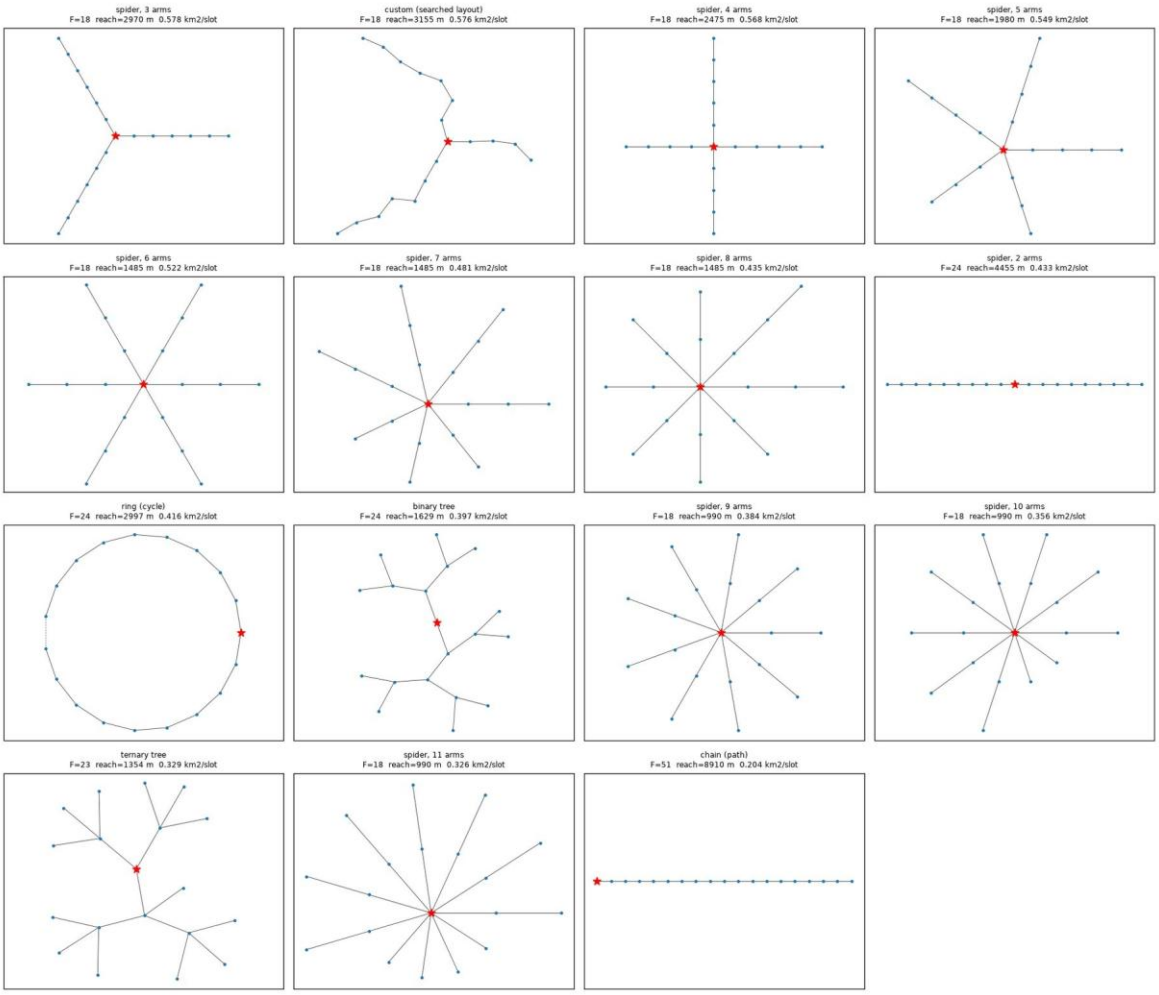


Figure 8.2: 19-radio topology comparison output.

Topology	Frame (slots)	Reach (m)	Max hops	Area (km²)	Area/slot
Spider 3 arms	18	2,970	6	10.40	0.5778
Custom searched	18	3,155	7	10.38	0.5765
Spider 4 arms	18	2,475	5	10.22	0.5677
Spider 5 arms	18	1,980	4	9.88	0.5491
Spider 6 arms	18	1,485	3	9.39	0.5218
Spider 7 arms	18	1,485	3	8.66	0.4814
Spider 8 arms	18	1,485	3	7.82	0.4346
Spider 2 arms	24	4,455	9	10.40	0.4334
Ring	24	2,997	9	9.98	0.4160
Binary tree	24	1,629	4	9.53	0.3971
Spider 9 arms	18	990	2	6.91	0.3841
Spider 10 arms	18	990	2	6.40	0.3556
Ternary tree	23	1,354	3	7.56	0.3287

Spider 11 arms	18	990	2	5.87	0.3263
Chain	51	8,910	18	10.40	0.2039

Interpretation: the three-arm spider is the strongest choice when area per slot is the primary metric. It uses the minimum 18-slot frame and reports 0.5778 km²/slot. The chain extends 8.910 km but requires 51 slots and has the lowest area efficiency in this set. The 9-, 10- and 11-arm spiders reduce maximum hops to two, but their covered area and area-per-slot values are lower.

8.2 Results for the 476-Radio Baseline

The 476-radio summary reports 13 feasible candidate topologies. Eight spider variants achieve the 475-slot lower bound. The chain has the greatest reach but also the largest frame length and maximum hop count.

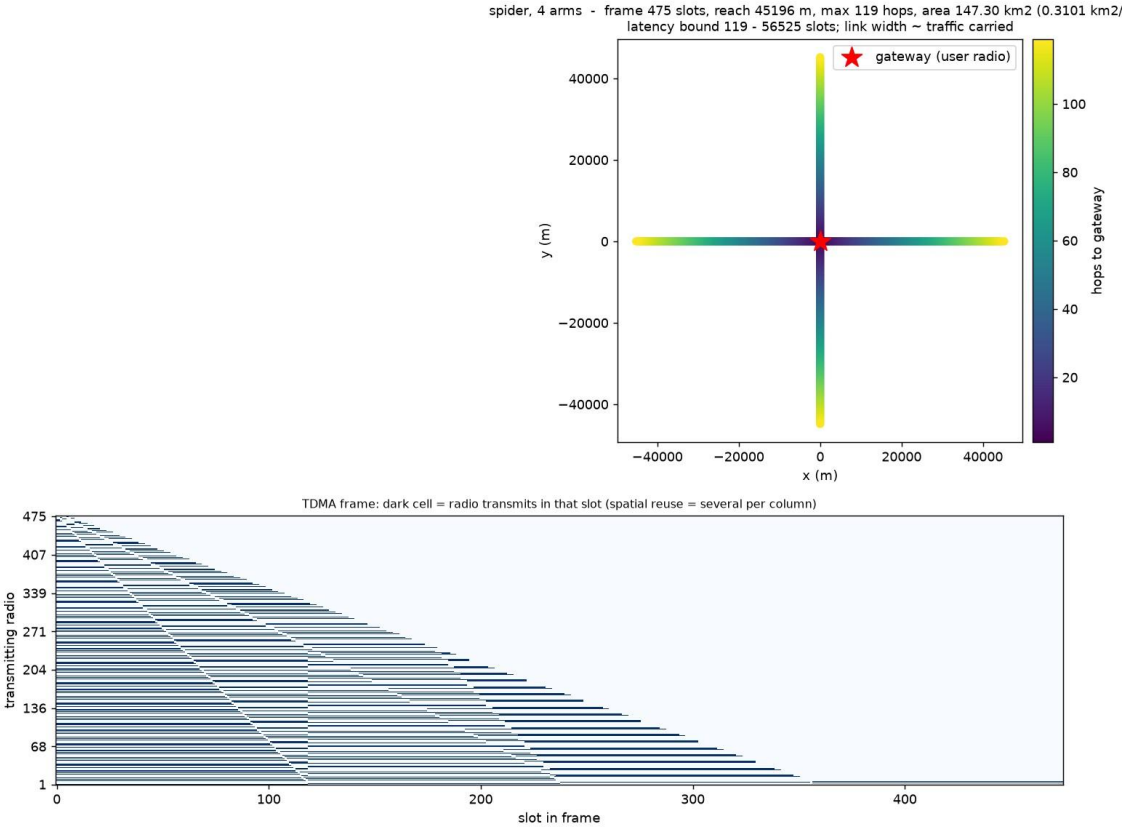


Figure 8.3: Best topology output for the 476-radio baseline.

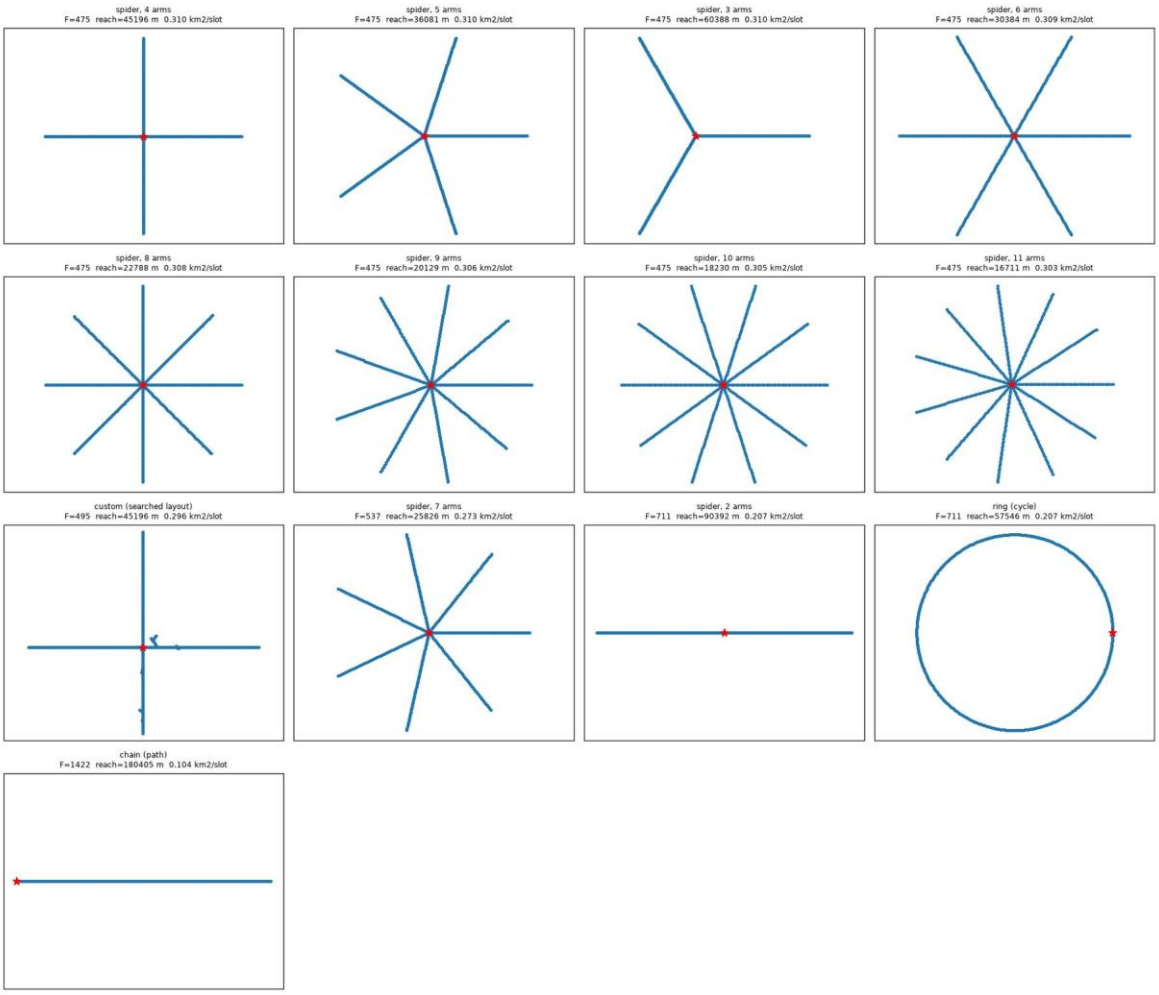


Figure 8.4: 476-radio topology comparison output.

Topology	Frame (slots)	Reach (m)	Max hops	Area (km²)	Area/slot
Spider 4 arms	475	45,196	119	147.30	0.3101
Spider 5 arms	475	36,081	95	147.21	0.3099
Spider 3 arms	475	60,388	159	147.05	0.3096
Spider 6 arms	475	30,384	80	146.93	0.3093
Spider 8 arms	475	22,788	60	146.39	0.3082
Spider 9 arms	475	20,129	53	145.40	0.3061
Spider 10 arms	475	18,230	48	144.71	0.3046
Spider 11 arms	475	16,711	44	143.79	0.3027
Custom searched	495	45,196	119	146.29	0.2955
Spider 7 arms	537	25,826	68	146.41	0.2726
Spider 2 arms	711	90,392	238	147.37	0.2073
Ring	711	57,546	238	147.20	0.2070
Chain	1,422	180,405	475	147.37	0.1036

Interpretation: the four-arm spider has the highest reported area-per-slot efficiency (0.3101 km²/slot) among the feasible candidates. The three-arm spider also reaches the lower-bound frame length and extends 60.388 km, but its maximum hop count is 159. The eleven-arm spider has the smallest maximum hop count among the lowerbound spider candidates (44), but its area-per-slot value is lower (0.3027 km²/slot). **8.3 Objective-Wise Selection**

Objective	Selected candidate / result	Meaning
Frame length	Multiple spider layouts: 475 slots	The lower-bound frame is achieved by several candidates
Latency / hop depth	Spider 11 arms: 44 max hops	Lowest maximum hop count among the listed 475-slot spider candidates
Reach	Chain: 180.405 km	Largest reported reach, with 1,422 slots and 475 max hops
Covered area	Spider 2 arms: 147.37 km²	Largest reported area; frame length is 711 slots
Area per slot	Spider 4 arms: 0.3101 km²/slot	Best reported coverage efficiency

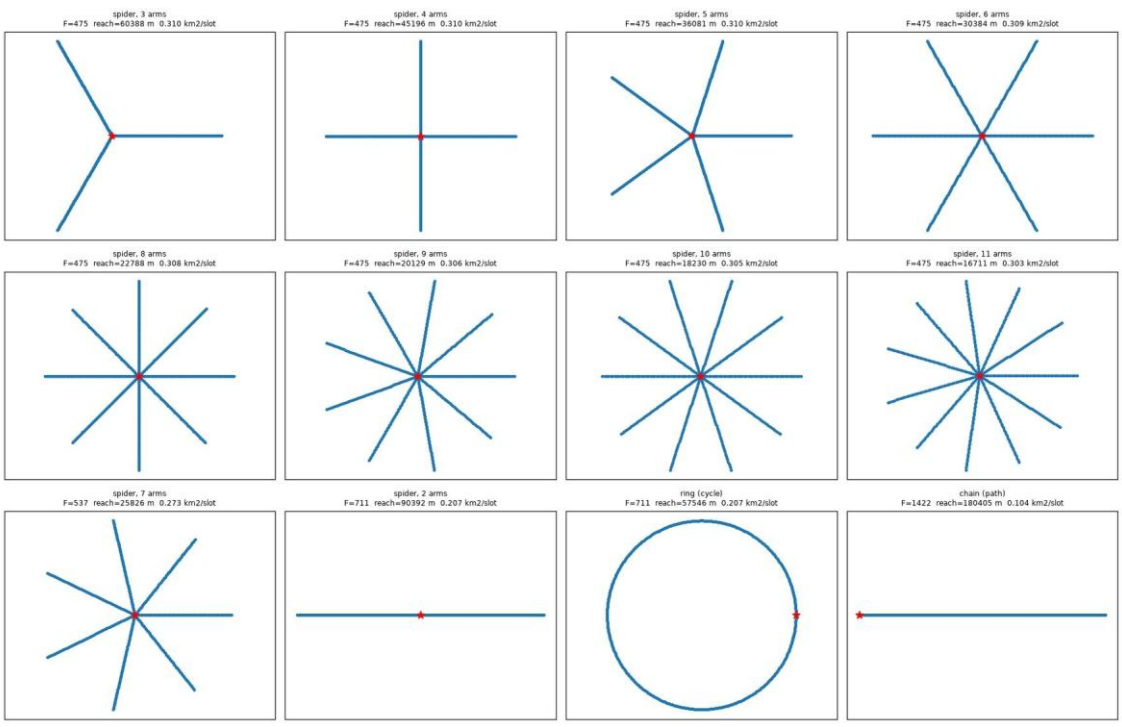


Figure 8.5: Comparison output from the frame objective rerun.

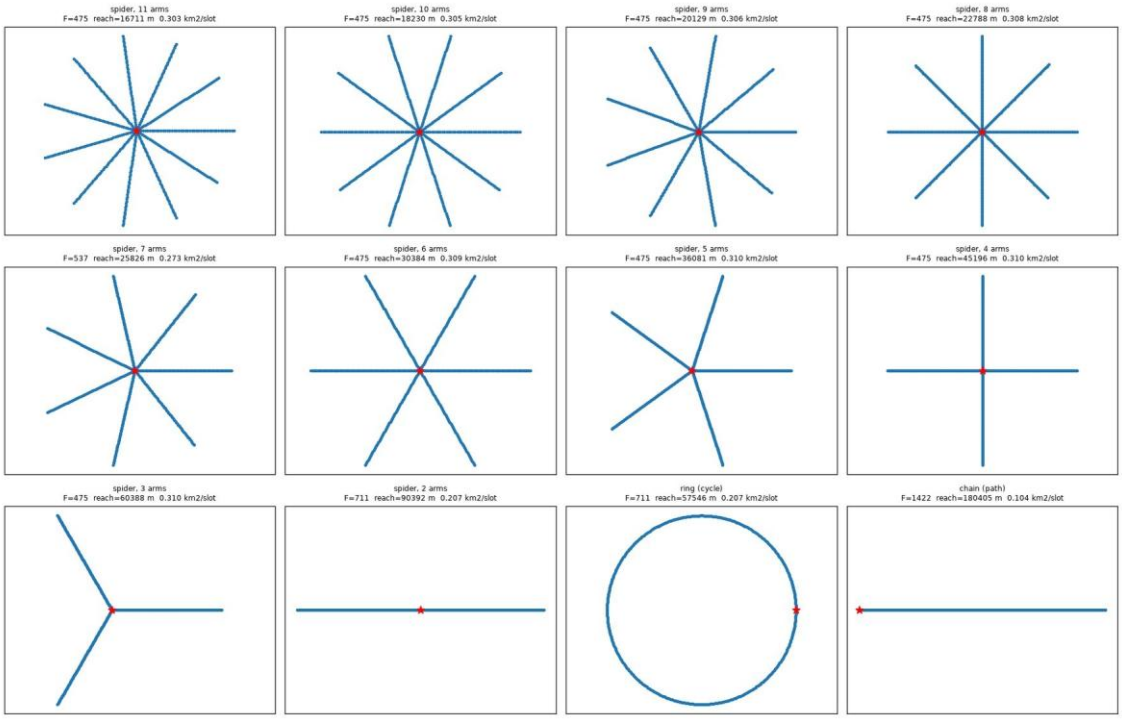


Figure 8.6: Comparison output from the latency objective rerun.

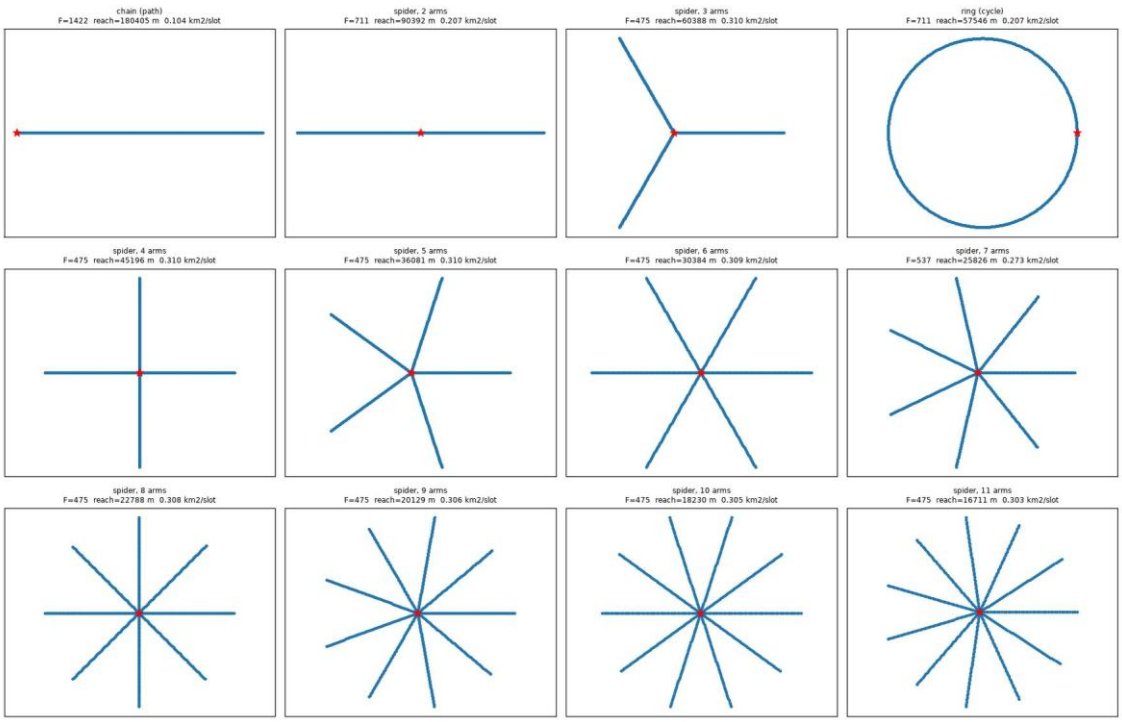


Figure 8.7: Comparison output from the reach objective rerun.

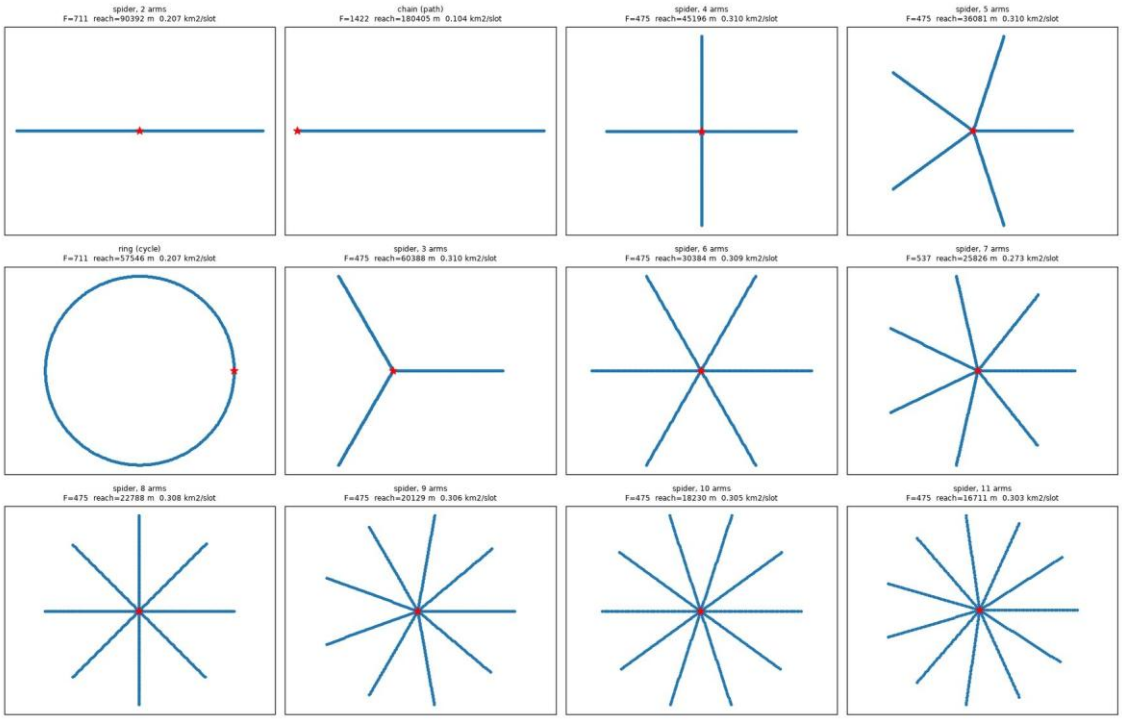


Figure 8.8: Comparison output from the area objective rerun.

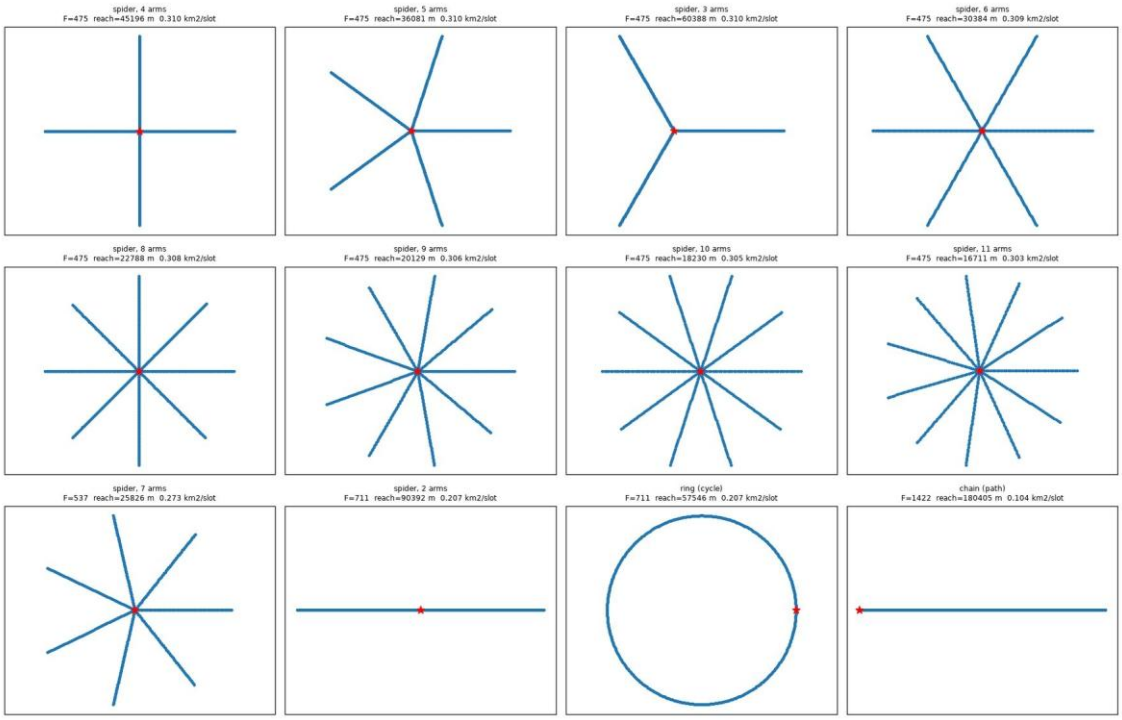


Figure 8.9: Comparison output from the efficiency objective rerun.

8.4 Custom Search Result

The custom searched 476-radio layout has a 495-slot frame, 45.196 km reach, 119 maximum hops, 146.29 km² reported area and 0.2955 km²/slot efficiency. It is feasible

in the summary, but it does not outperform the four-arm spider on area-perslot efficiency. The run used 2,000 proposals and is not sufficient to establish global optimality.

8.5 Results for the 50-Radio Scalability Case

The archive includes one additional scalability run with 50 radios and a 422 m range. The frame lower bound is 49 slots because there are 49 non-gateway radios. The three-arm spider is reported as the best area-per-slot candidate with a 49-slot frame, 6,457 m reach, 17 maximum hops, 15.72 km² area and 0.3209 km²/slot.

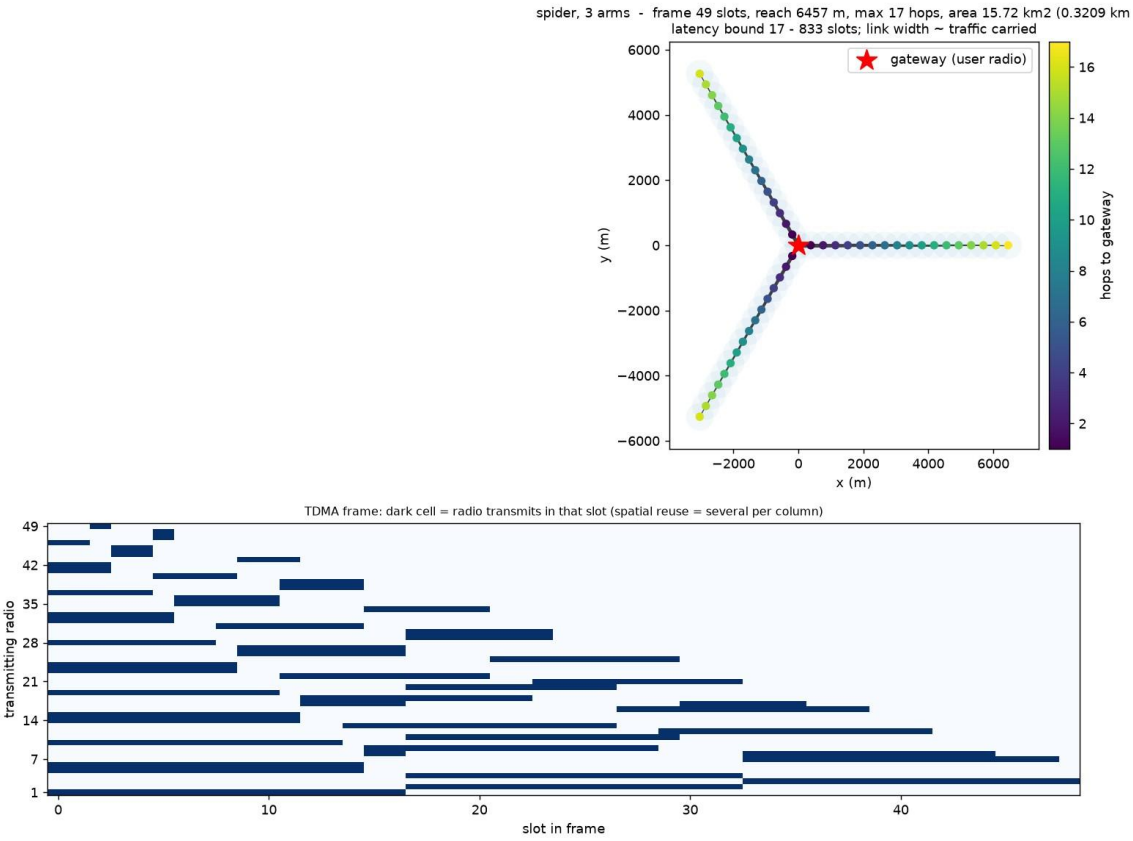


Figure 8.10: Best topology output for the 50-radio scalability run.

The chain candidate reaches 18.610 km but requires a 144-slot frame and has 49 maximum hops. This single intermediate run is useful as an additional observation, but it is not a complete scalability curve.

Case	Radios	Frame lower bound	Best area/slot candidate	Best area/slot	Farthest reach candidate
19-radio	19	18	Spider 3 arms	0.5778	Chain: 8.910 km
50-radio	50	49	Spider 3 arms	0.3209	Chain: 18.610 km
476-radio	476	475	Spider 4 arms	0.3101	Chain: 180.405 km

8.6 Feasibility Rejections

For the 476-radio case, the 12-arm spider was rejected because a reported pair was separated by 197 m, below the configured 211 m minimum. The star candidate had a reported pair separation of 5 m. The binary and ternary tree variants were also rejected in the 476-radio configuration according to the summary. For the 19-radio case, the 12-arm spider and star were rejected because reported pair distances were 256 m and 172 m respectively, both below the 275 m minimum. These are specific candidate outcomes.

8.7 Latency and Scheduling Interpretation

The four-arm spider's schedule-aware worst-case latency is reported as 28,498 slots for the deepest radio under the one-packet and no-queueing assumptions. The topology summary also contains a hop-based latency upper-bound field. The two should not be conflated: one is a schedule-aware calculation and the other is the topology summary's bound. Since a slot duration is not specified, neither value is converted to seconds.

9. DISCUSSION

9.1 No Universal Best Topology

The results demonstrate a multi-objective trade-off. The chain maximizes reach but incurs the largest frame and hop depth. Spider topologies offer a more balanced compromise. A four-arm spider is best by area per slot in the 476-radio case, while a three-arm spider is best by area per slot in the 19- and 50-radio cases.

9.2 Frame-Length Lower Bound

The achievement of 475 slots by several 476-radio candidates is significant because it matches the lower bound of 475 source packets per frame under the stated singlegateway demand model. This shows that the scheduler can attain the theoretical minimum for those candidate structures in the tested configuration.

9.3 Effect of Arm Count

Increasing the number of spider arms generally reduces maximum hop count in the listed 476-radio candidates, but it also reduces reach and area-per-slot efficiency. The arm count therefore provides a practical design control for balancing network depth against spatial footprint.

9.4 Scale Comparison

The 19-, 50- and 476-radio cases suggest that the preferred spider arm count may vary with network size. However, the radio range differs between the 19-radio case and the other two cases, and only one intermediate-scale point is available. A controlled scalability conclusion requires additional runs with consistent radio parameters.

9.5 Engineering Implications

For applications where predictable frame length and efficient spatial coverage are important, a spider topology is a strong candidate. If long geographical reach is dominant and additional forwarding delay is acceptable, a chain may be considered. If low routing depth is the priority, a higher-arm spider may be more suitable. Final selection should also consider physical radio behaviour, energy, reliability, traffic demand and deployment constraints.

10. LIMITATIONS AND FUTURE WORK

10.1 Evidence Limitations

- Only one additional intermediate scalability run (50 radios) is present in the archive.
- The planned 100, 200, 300, 400 and 600 radio scalability runs are absent.
- No completed traffic-load sweep is present.
- No completed interference-range sweep is present.
- No completed minimum-separation sweep is present.
- No completed link-distance-factor sweep is present.
- No completed multi-seed statistical evaluation is present.
- The report does not include physical testbed or over-the-air validation.
- The duration of a TDMA slot is not specified, so latency is retained in slots.

10.2 Future Work

- Complete a controlled scalability sweep at 50, 100, 200, 300, 400, 476 and 600 radios using consistent range and constraint settings.
- Run traffic-load experiments with multiple packets per node and report queueing effects.
- Vary interference range and compare the resulting conflict graph and frame length.
- Sweep minimum separation and link-distance factor to quantify feasibility sensitivity.
- Run multiple random seeds and report mean, standard deviation and confidence intervals.

- Measure energy consumption, packet delivery ratio, throughput and end-to-end delay.
- Validate selected layouts using a network simulator or a physical wireless testbed.
- Convert slot-based latency into time only after defining the slot duration and timing overhead.

10.3 Recommended Validation Metrics

Metric	Recommended measurement
Packet delivery ratio	Successfully delivered packets / generated packets
Throughput	Payload bits delivered per second
End-to-end delay	Receive timestamp minus generation timestamp
Energy per delivered packet	Total network energy / delivered packets
Schedule utilization	Used slots / total available slots
Reliability	Repeated-run delivery and connectivity statistics

11. CONCLUSION

This project developed and evaluated a software-based workflow for comparing TDMA wireless network topologies. The supplied results show that topology selection has a direct impact on frame length, hop count, reach and area efficiency. In the 476-radio case, multiple spider candidates attain the 475-slot theoretical lower bound. The four-arm spider provides the highest reported area-per-slot efficiency at 0.3101 km²/slot, while the chain provides the largest reach at 180.405 km with a much larger 1,422-slot frame. In the 19-radio case, the three-arm spider achieves 0.5778 km²/slot and the 18-slot lower bound. The 50-radio run provides one additional intermediate observation, with the three-arm spider reporting 0.3209 km²/slot.

The central conclusion is that topology optimization is inherently multi-objective. A design should be selected according to the dominant system requirement rather than a single metric. The current evidence supports comparative conclusions for the completed cases, while broader claims about scalability, traffic sensitivity and robustness require the missing experiment sweeps and physical validation.

12. CASE STUDY: N×N TDMA OPTIMIZER

This chapter documents the N×N (N-radio) TDMA optimizer as a stand-alone case study. It is the second-stage software that grew out of the fixed 4×4 planner: instead of one grid size, it accepts any number of radios N and a radio range R, generates many arrangements, schedules each one and reports which arrangement is best. All figures are outputs of this software.

12.1 Problem and Approach

Node 0 is the user's radio (the gateway). Every other radio sends one packet per frame to it, forwarding for its neighbours. The optimizer builds a routing tree to the gateway, converts it into a conflict graph (radios that cannot transmit in the same slot), and colours that graph to obtain the TDMA slots. Its main quantity is the frame length F, the number of slots before the schedule repeats.

Item	Setting
Radio range R	Value given by the user (100 m in the cases below)
Neighbour spacing	0.9 R, so every link keeps a 10% margin
Interference range	1.5 R (disk model); SINR model tested in Section 12.8
Traffic	Convergecast: every radio and user reports once per frame
Slot assignment	Weighted greedy colouring; also a distributed randomized version
Frame floor	The gateway receives one packet per slot, so $F \geq N - 1$ (plus attached users)

12.2 Ranking Topologies for a Given N and Range

Case A: 10 radios, 100 m, no users. The frame floor is 9 slots. The star and the 3and 4-arm spiders reach this floor but cover only 90 to 270 m. The chain reaches 810 m but needs 24 slots. The 2-arm spider offers the best balance: 450 m in 12 slots, or 37.5 m of reach per slot.

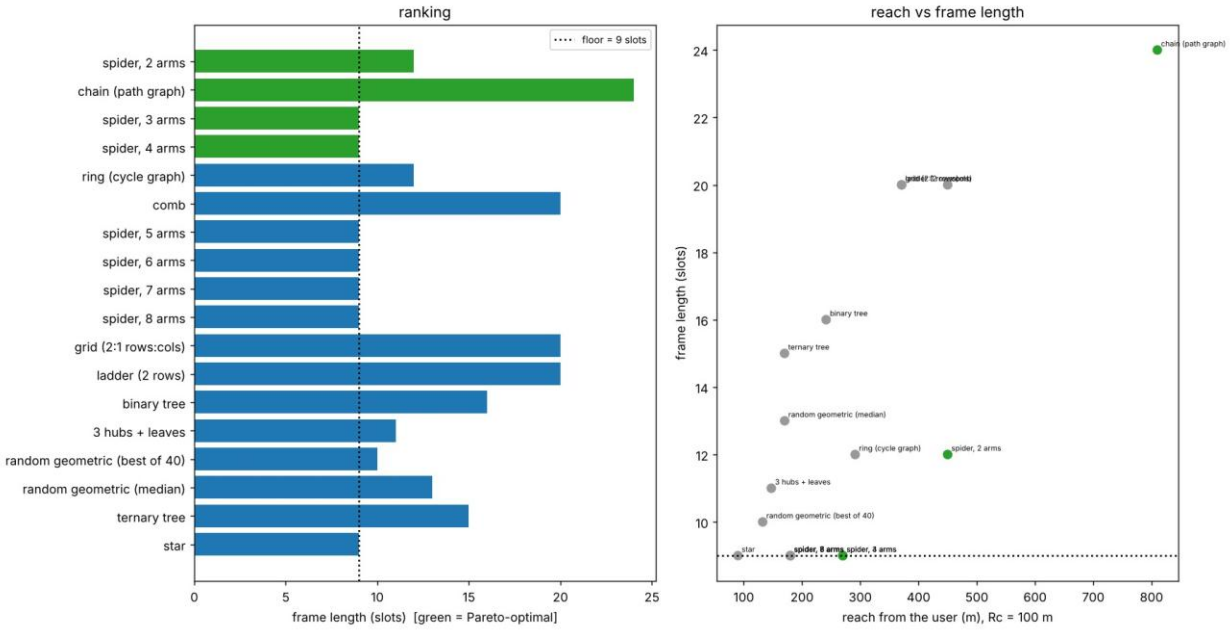


Figure 12.1: Case A (10 radios): frame length of every topology (left) and reach versus frame length (right).

Left: green bars are Pareto-optimal, meaning no other topology is both shorter and farther-reaching. Right: moving right gains reach, moving up costs frame length.

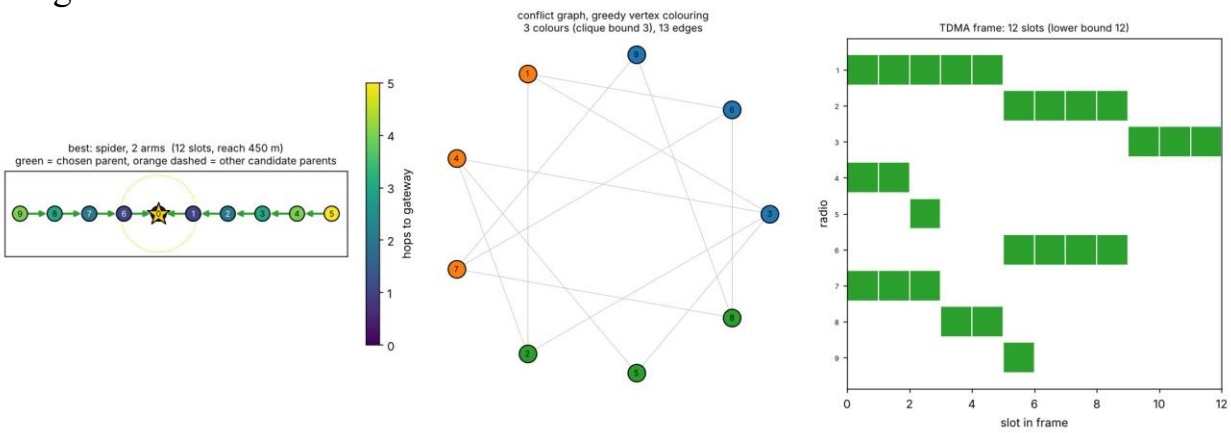


Figure 12.2: Case A: best balance, the 2-arm spider, with its conflict graph and TDMA frame.

Left: two arms of five radios on either side of the user. Middle: the conflict graph needs 3 colours. Right: the 12-slot frame equals its lower bound, so no shorter schedule exists for this arrangement.

Case B: 18 radios, 100 m, 40 attached users, target distance 800 m. The gateway must receive 57 packets per frame. Only four candidates reach 800 m: the 2-arm spider (89 slots, 810 m, 9 hops), the search-optimised layout (89 slots, 810 m), the comb (146 slots) and the chain (158 slots, 1,530 m). The star, ring, trees and spiders with three or more arms are excluded because they stop short of 800 m.

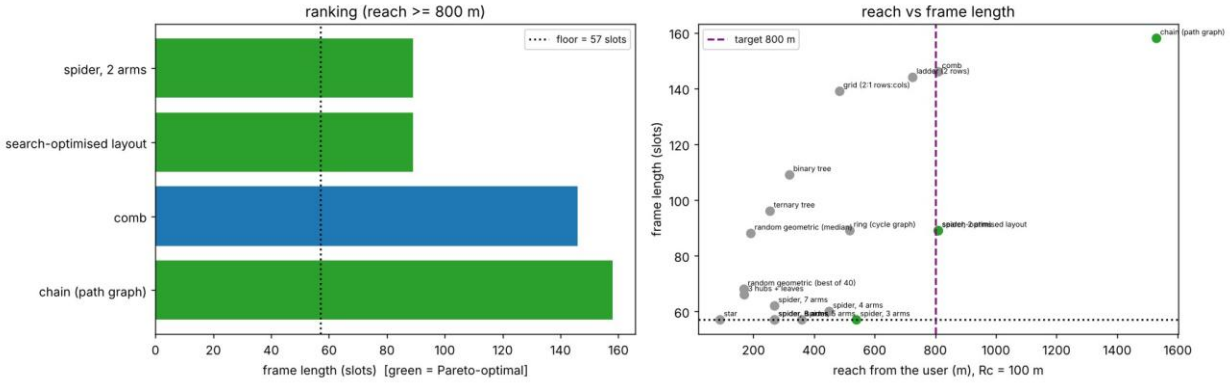


Figure 12.3: Case B (18 radios, 40 users): only topologies that reach 800 m are ranked.

The dotted line is the 57-slot floor. Dashed purple marks the 800 m target; points left of it fail the requirement.

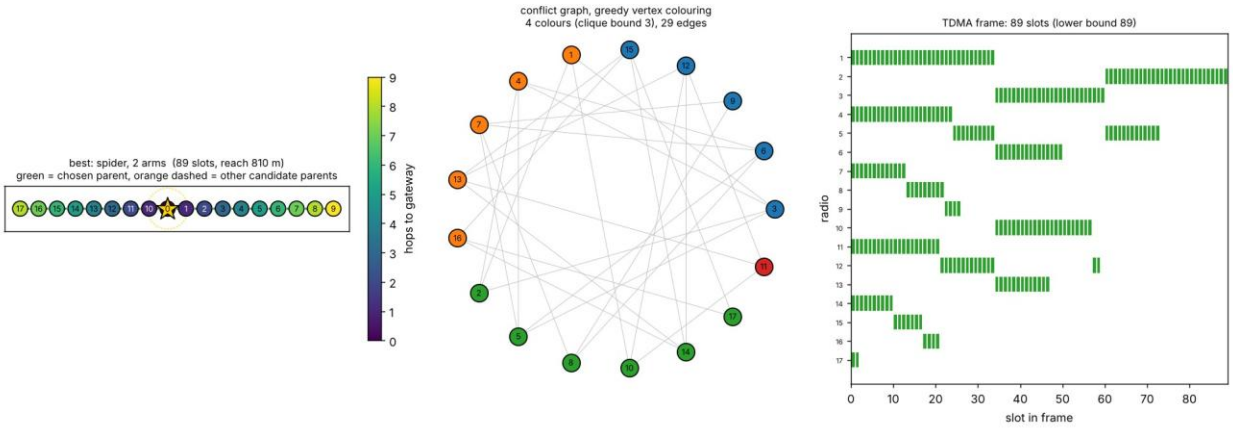


Figure 12.4: Case B: the 2-arm spider, its conflict graph (4 colours, 29 edges) and the 89-slot frame.

The frame is 32 slots above the 57-slot floor because radios near the gateway conflict with each other. Far-apart radios share slots, which is the spatial reuse visible in the right panel.

Case	Floor	Best choice	Frame	Reach	Hops
10 radios, no users	9	2-arm spider	12	450 m	5
18 radios, 40 users, ≥ 800 m	57	2-arm spider	89	810 m	9

12.3 Graph-Theory View of All Arrangements

Each arrangement is a graph: radios are vertices, usable links are edges, and the green arrows are the routing tree. The figures below compare the named patterns and then every arrangement the search tried.

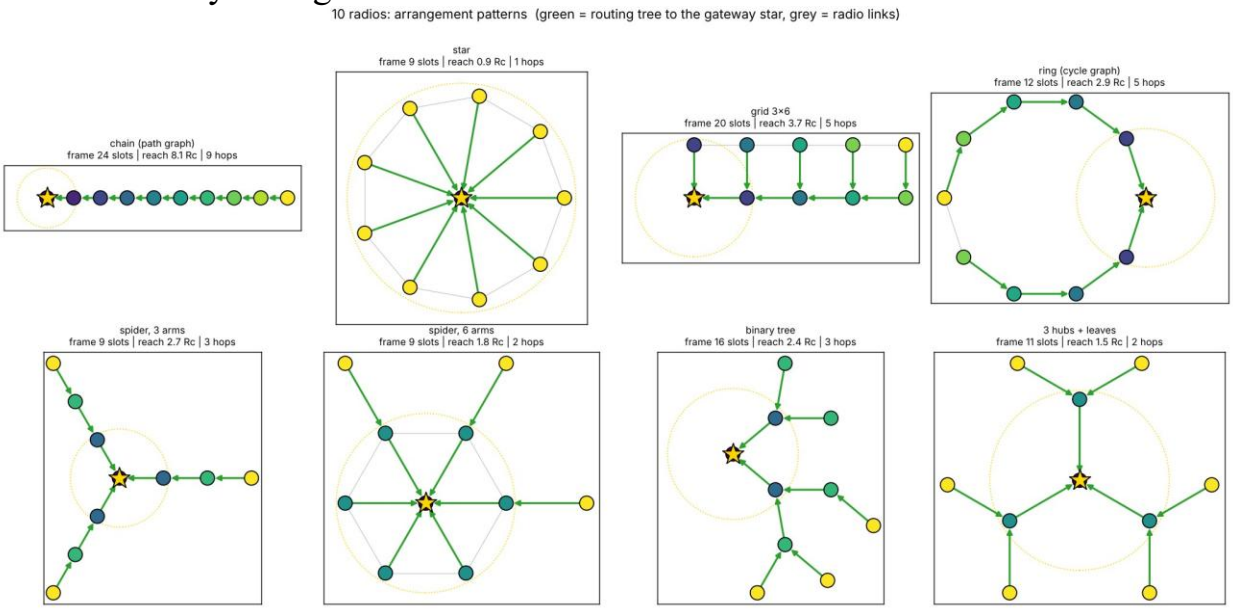


Figure 12.5: Case A: the eight named patterns (green = routing tree, colour = hops to the gateway).

Star: 9 slots, reach 0.9 R. Chain: 24 slots, 8.1 R. 3-arm spider: 9 slots, 2.7 R. Ring: 12 slots, 2.9 R. Grid 3×6: 20 slots, 3.7 R. Binary tree: 16 slots, 2.4 R. Hubs and leaves: 11 slots, 1.5 R.

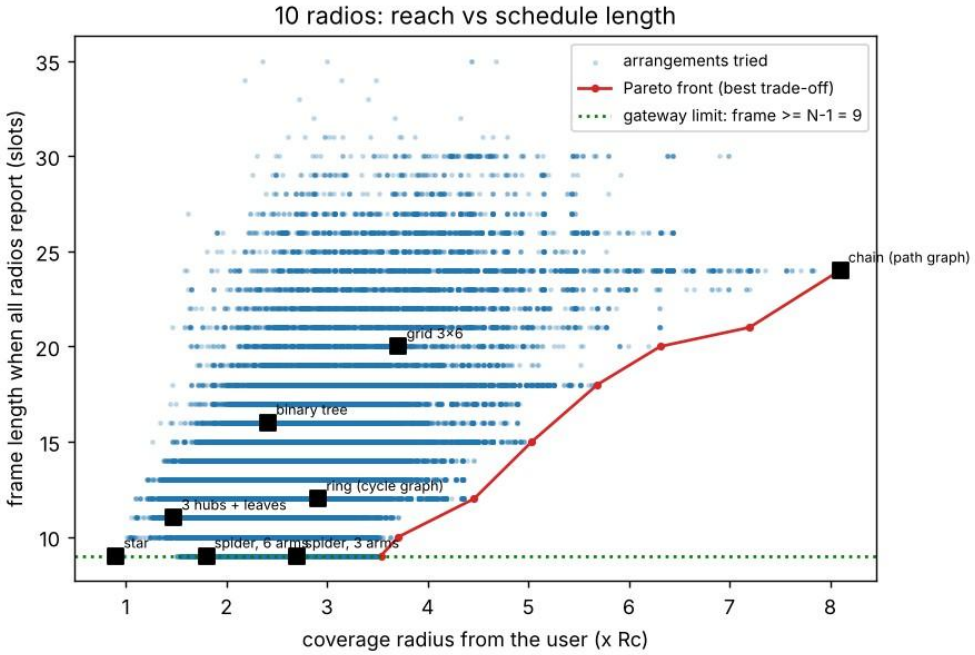


Figure 12.6: Case A: every arrangement tried (blue), the Pareto front (red) and the named patterns (black squares).

The front rises from 9 slots at about 3.5 R to 24 slots at 8.1 R. Several searched layouts beat the ring, grid and trees, while the 3-arm spider sits close to the front.

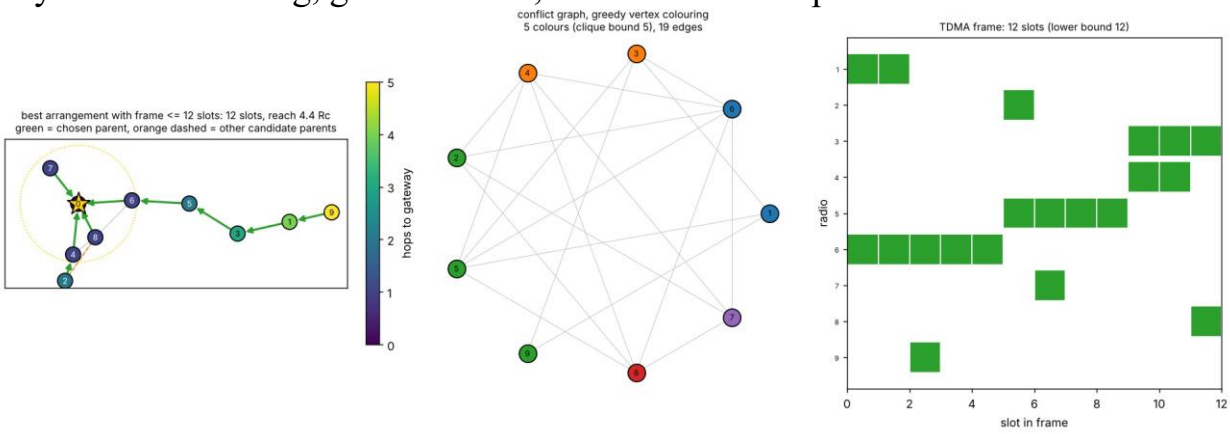


Figure 12.7: Case A: best arrangement found by the search for a frame of at most 12 slots.

It reaches 4.4 R in 12 slots, matching the 2-arm spider (4.5 R), so for 10 radios a regular pattern is already near-optimal. The conflict graph here needs 5 colours.

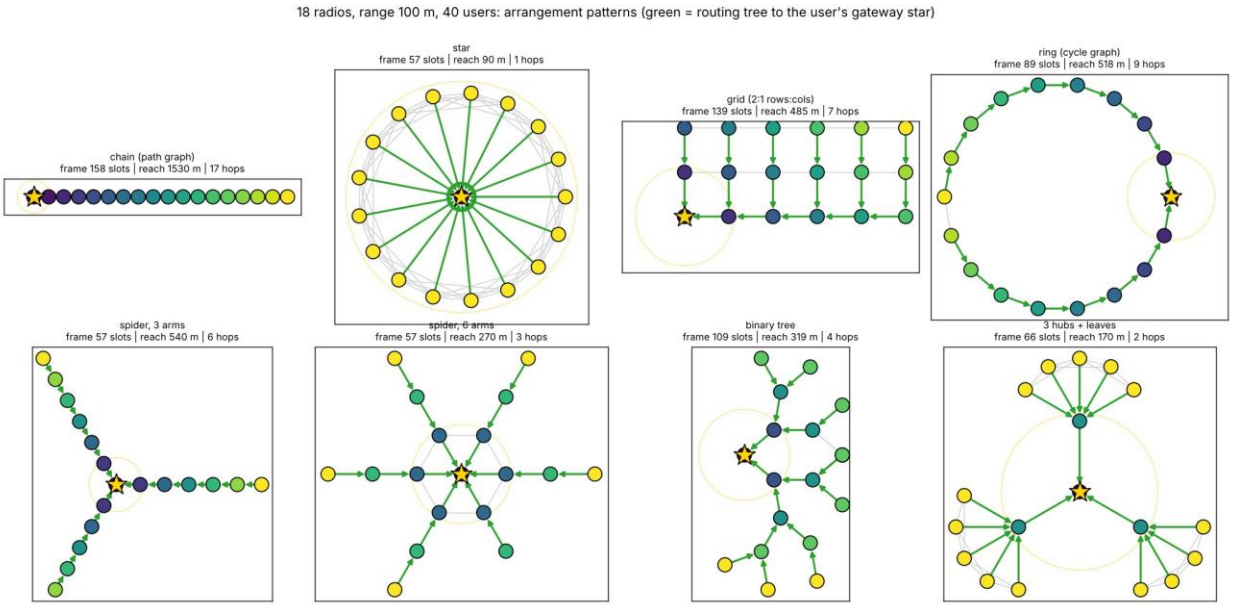


Figure 12.8: Case B: the eight named patterns for 18 radios and 40 users.

Star: 57 slots, 90 m. Chain: 158 slots, 1,530 m. Grid: 139 slots, 485 m. Ring: 89 slots, 518 m. 3-arm spider: 57 slots, 540 m. Binary tree: 109 slots, 319 m. Hubs and leaves: 66 slots, 170 m.

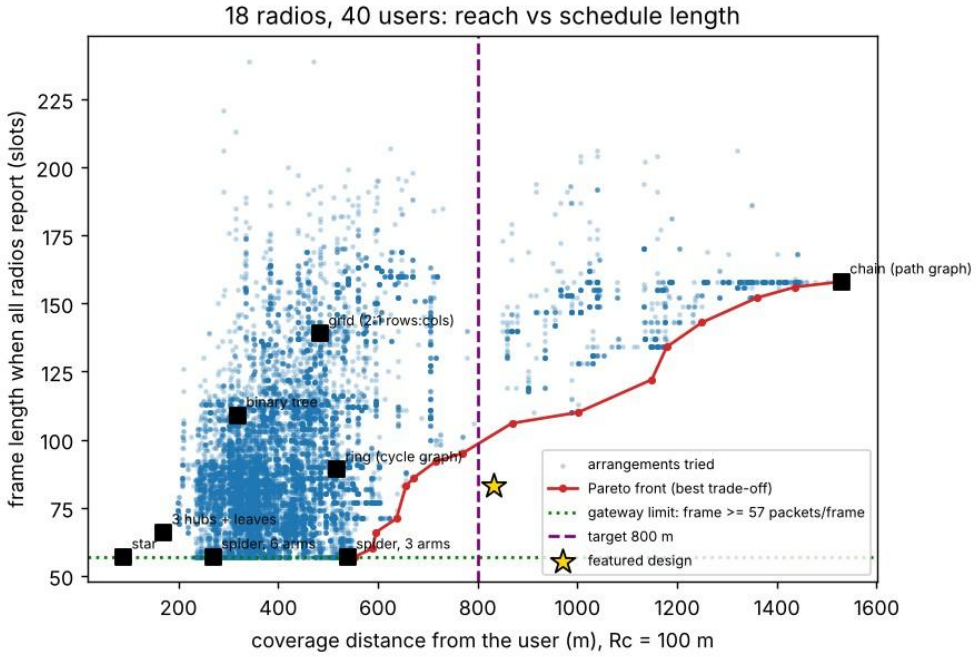


Figure 12.9: Case B: reach versus frame length; the star marks the featured design.

The featured design lies to the right of the 800 m target and well below the cloud of random layouts, so it needs a short frame for its reach.

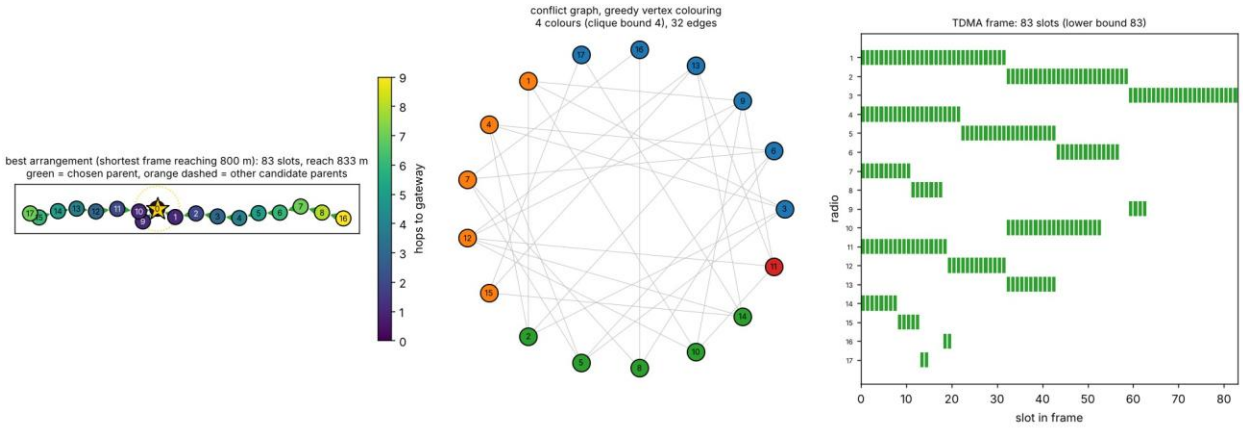


Figure 12.10: Case B: featured design, the shortest frame reaching 800 m.

The search found a slightly bent chain with 83 slots and 833 m reach, better than the 2-arm spider (89 slots, 810 m). It is the one case where the search beat every named pattern; its 83-slot frame equals its lower bound.

12.4 TDMA versus Random Access

To justify scheduling, the planner was compared with slotted ALOHA and slotted CSMA on 20 radios (5 random topologies, 8,000 slots each). With the same frame length, random slot choice collides in 65.9% of transmissions; the planner collides in 0%.

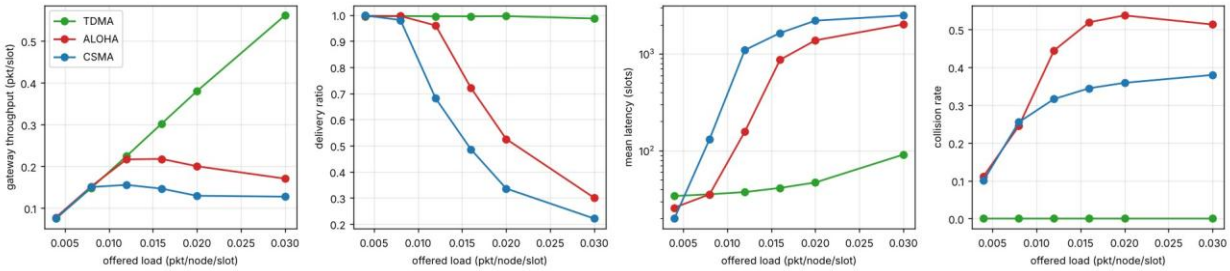


Figure 12.11: Throughput, delivery ratio, latency and collision rate for TDMA, ALOHA and CSMA as the offered load grows.

At a load of 0.03 packets per node per slot, TDMA delivers 98.6% at 0.561 packets per slot with a 91-slot mean latency. ALOHA falls to 30% delivery and CSMA to 22%, with latencies above 2,000 slots.

12.5 Coverage Distance and Service Level

Coverage can be bought only at the cost of rate and delay. For a chain at 0.9 R spacing with 10 ms slots, a single flow always needs 3 slots, but when all radios report, the frame grows by 3 slots for every added relay.

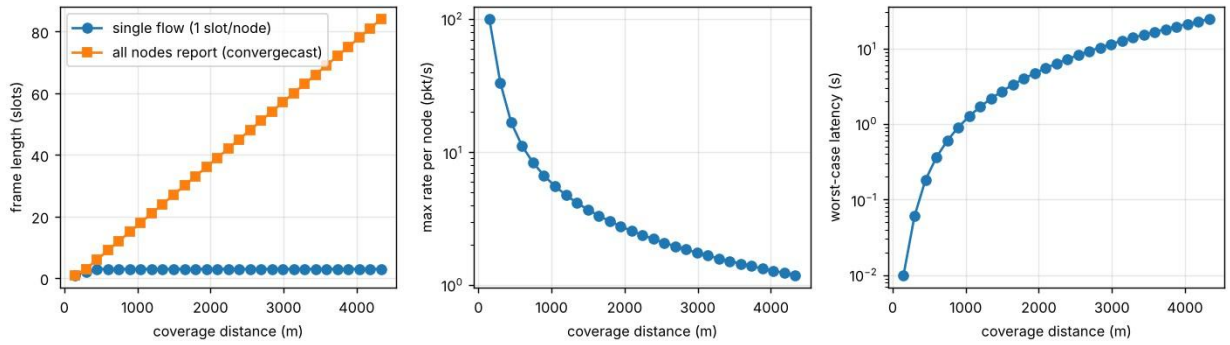


Figure 12.12: Coverage distance versus frame length, maximum rate per node and worst-case latency.

Six radios (747 m) allow 8.3 packets/s per node with 0.6 s worst latency. Thirty radios (4,334 m) allow 1.19 packets/s with 24.4 s. A service of 1 packet/s and 5 s latency limits coverage to 1,943 m (13 hops).

12.6 Scalability and Optimality

From 10 to 200 radios the greedy frame stays within 2% of the lower bound (ratio 1.000 to 1.019), planning takes 0.06 to 16.5 ms, and the distributed randomized version converges in about three rounds.

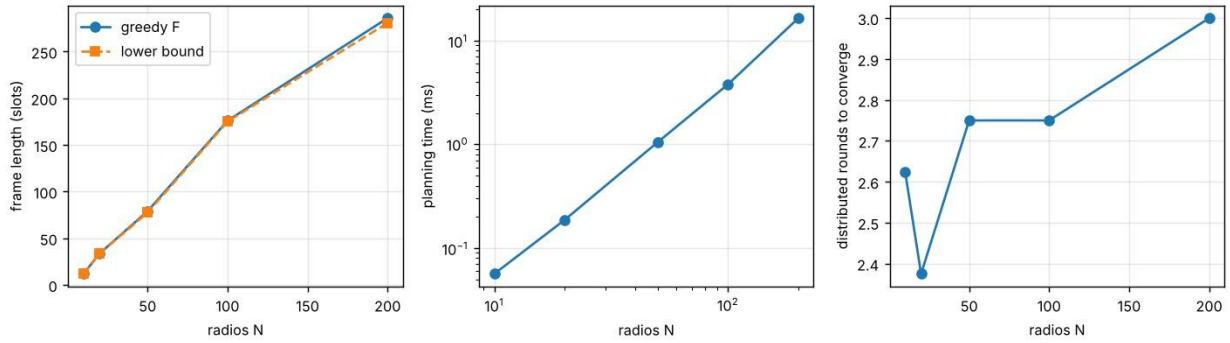


Figure 12.13: Frame length against its lower bound, planning time and distributed convergence for 10 to 200 radios.

Spatial reuse grows from 1.18 to 4.03 simultaneous transmissions per slot as N increases. The schedule was conflict-free in every run.

An exact DSATUR solver checked the greedy result on 30 random instances for each size. Greedy equalled the optimum on every random tree (N = 6 to 20). On 4×4, 5×5 and 6×6 grids it needed one extra slot.

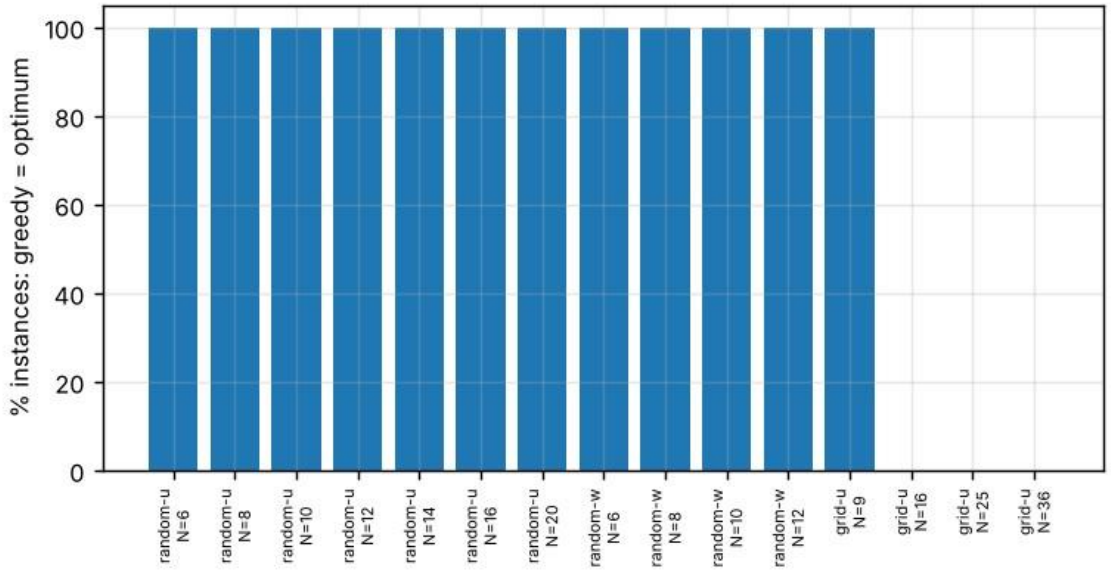


Figure 12.14: Share of instances where the greedy schedule equals the proven optimum.

Random families are all 100%. The grids with N = 16, 25 and 36 show 0%, a gap of exactly one slot, which is the honest limit of the greedy method.

12.7 Dynamic Joins and Leaves

When a radio joins or leaves, only the affected radios are rescheduled.

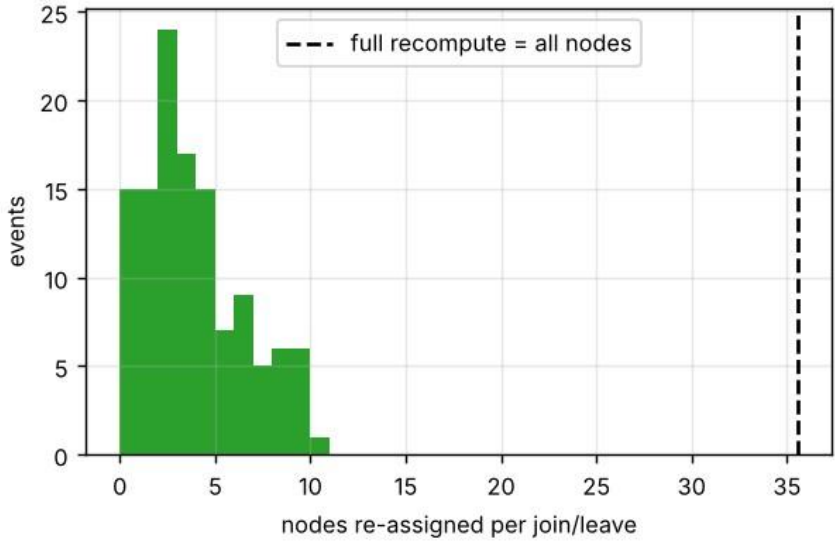


Figure 12.15: Radios re-assigned per join or leave event (120 events).

The schedule stayed valid after every event. On average 3.4 radios changed slots (maximum 10), about 9.5% of a network of roughly 36, against 100% for a full recompute.

12.8 Realistic Interference (SINR and Shadowing)

The disk model ignores the sum of many weak interferers. With 30 radios and 12 random seeds, an SINR model with log-normal shadowing (σ = 0, 4 and 8 dB) was applied to the disk-model schedules.

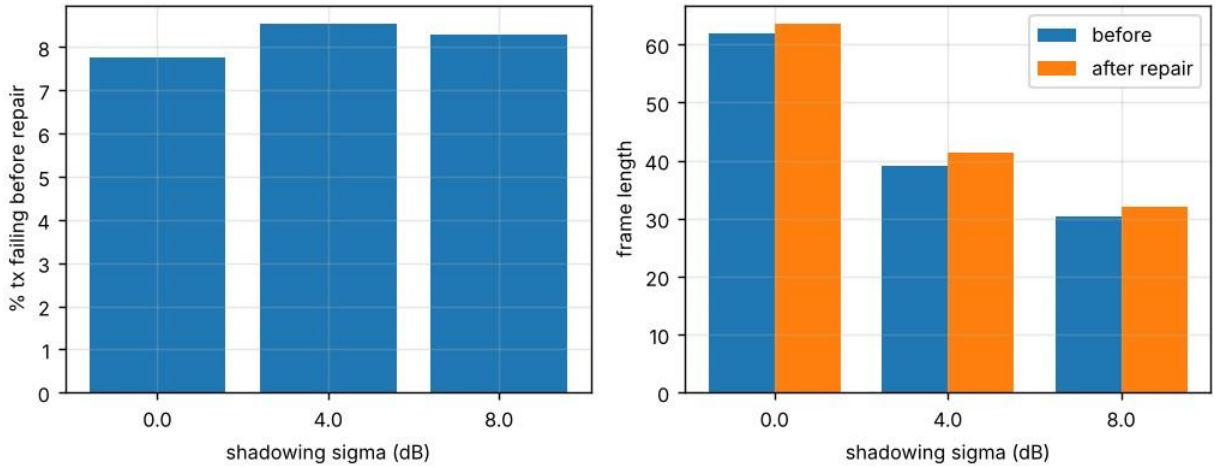


Figure 12.16: Transmissions failing before repair, and frame length before and after repair.

Between 7.8% and 8.5% of transmissions failed. A repair loop that moves failing transmissions removed all failures in 3.8 to 5.6 iterations and added only 1.6 to 2.1 slots.

12.9 Summary of the Case Study

Question	Result
Best balance (10 radios)	2-arm spider, 12 slots, 450 m
Best for 800 m (18 radios, 40 users)	Searched bent chain, 83 slots, 833 m (2-arm spider: 89 slots, 810 m)
Longest reach	Chain, at the cost of the longest frame
TDMA against random access	0% collisions and 98.6% delivery against 30% and 22% at the highest load
Scalability	Frame within 2% of the bound up to 200 radios, planning under 17 ms
Greedy against optimum	Optimal on random trees; one slot above on grids

Limitations. Reception uses a disk model with a fixed interference range, and only the SINR experiment tests aggregated interference. Slots are assigned greedily, so grids can be one slot above the optimum. Latency is reported as a hop-based bound. Results are simulated; no hardware test has been completed.

REFERENCES

[1] T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed., Prentice Hall, 2002.

[2] A. S. Tanenbaum and D. J. Wetherall, Computer Networks, 5th ed., Pearson, 2011.

[3] W. Stallings, Wireless Communications and Networks, 2nd ed., Pearson, 2005.

[4] Project experimental archive: TDMA_Complete_Experimental_Results.zip, supplied result summaries, JSON files, CSV layouts, run logs and PNG outputs, accessed 8 November 2026.

Note: The references above provide general background. Numerical findings in this report are taken from the supplied project outputs, not from the textbooks.

APPENDIX A – EXPERIMENTAL EVIDENCE

The supplied archive contains the following principal result groups:

- n19_r550: 19-radio baseline summary, candidate layouts and plots.
- n476_r422: 476-radio baseline summary, candidate layouts and plots.
- objective_frame: 476-radio frame-oriented ranking rerun.
- objective_latency: 476-radio latency-oriented ranking rerun.
- objective_reach: 476-radio reach-oriented ranking rerun.
- objective_area: 476-radio area-oriented ranking rerun.
- objective_efficiency: 476-radio area-per-slot ranking rerun.
- scalability/n50/n50_r422: one 50-radio scalability result.

APPENDIX B – REPRODUCTION GUIDE

The project was previously run in Ubuntu 24.04 under WSL2 with Python 3.12. The following is a generic environment setup sequence; use the exact command-line arguments and source version associated with the original run when reproducing results.

cd ~/tdma_topology_optimizer python3 -m

venv .venv source .venv/bin/activate

python -m pip install --upgrade pip python

tdma_topology_optimizer.py --help

The help output should be consulted for the exact accepted parameters. Do not assume a parameter value or run category was tested unless its output is present in the archive.

APPENDIX C – RESULT INTERPRETATION

A lower frame length is desirable for compact scheduling. A lower maximum hop count is desirable for reducing routing depth. Greater reach is desirable for spatial extension. Greater area-per-slot efficiency is desirable when coverage must be balanced against schedule length. These objectives can conflict; the result tables should be interpreted as a decision aid rather than a single score.